Ministry of Earth Sciences (MoES)
India Meteorological Department (IMD)

Standard Operating Procedure: Doppler Weather Radar (DWR) Operations

1. Introduction to Doppler Weather Radars

Doppler Weather Radars (DWR) are critical tools used by the IMD to monitor severe weather events. Unlike conventional radars that only track the intensity of precipitation, Doppler radars utilize the Doppler effect to determine the velocity of precipitation particles (hydrometeors) in the atmosphere. This dual capability allows meteorologists to identify wind patterns within storm systems, providing early warnings for thunderstorms, squalls, and tropical cyclones.

2. Operational Calibration and Maintenance

For optimal performance, DWR systems must undergo daily calibration checks. The receiver sensitivity must be maintained within strict limits. If the noise figure exceeds the nominal threshold, the radome and antenna components must be inspected for moisture ingress. Furthermore, the pulse repetition frequency (PRF) must be adjusted according to the meteorological situation; a low PRF is suited for long-range reflectivity surveillance, whereas a high PRF is required for accurate velocity unfolding in severe localized convective storms.

3. Echo Interpretation and Cyclone Tracking

During tropical cyclone tracking, the DWR is pivotal in identifying the eye, the eyewall, and the spiral rainbands. A distinct hook echo on the reflectivity display often indicates the presence of a mesocyclone, which can spawn tornadoes. When issuing cyclone warnings, the DWR data must be cross-validated with INSAT-3D multispectral satellite imagery to confirm the structural integrity of the storm system and estimate the central pressure drop.